



Black Shark Collective

Electronics, Firmware & Configuration Guide

Applies to	Details
Device	Black Shark Collective with swappable grip and twist throttle
Firmware	HOTAS Collective firmware, dual PCF8575 (03_Collective.ino)
Configuration app	Tactical MFD Control Hub (Windows application, .exe)
Firmware updates	https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main
Companion document	Black Shark Collective – Assembly Instructions (inputs layout, BOM, schematic, mechanical build)

Use this guide after the mechanical assembly is finished. It covers the wiring and I2C addresses, how to verify the wiring, how to flash the firmware and how to configure the collective with the Tactical MFD Control Hub app (calibration, axis inversion and filters).

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1.- How the system works

The collective is split in two parts. The **base** (MCU housing and handle) holds the Arduino Pro Micro and the two axis sensors, which connect directly to analog pins. The **grip** (collective head) holds all the buttons, read by two PCF8575 expanders. The grip connects through a 4-pin 360° pogo connector (I2C and power), so it can be swapped and the twist throttle can rotate freely.

Location	Component	I2C address	Function
Base	Arduino Pro Micro (ATmega32U4)	—	Main controller: USB game controller + serial port
Base	SS49E hall sensor	—	Collective lever – Rx axis (pin A1)
Handle	SS49E hall sensor	—	Twist throttle – Ry axis (pin A2)
Handle	360° pogo connector (4-pin)	—	Links the grip to the base: VCC, GND, SDA, SCL
Grip	PCF8575 expander #1	0x20	Buttons 1–16
Grip	PCF8575 expander #2	0x21	Buttons 17–32

1.1.- Where your settings are stored

Calibration, axis inversion, filters and the jitter threshold are all stored in the Pro Micro's internal EEPROM, in the base. They survive power cycles and firmware updates. The grip only carries buttons, so you can swap grips without recalibrating.

Swapping grips

Unplug the USB cable before you disconnect or connect a grip. Plug it back in once the new grip is in place.

2.- Wiring and I2C addresses

2.1.- Base wiring

Pro Micro pin	Connects to
A1	Collective lever sensor (SS49E signal) – Rx axis
A2	Twist throttle sensor (SS49E signal) – Ry axis
Pin 2 (SDA)	Pogo connector – SDA
Pin 3 (SCL)	Pogo connector – SCL

Pro Micro pin	Connects to
VCC (5 V)	Pogo connector VCC and both sensors
GND	Pogo connector GND and both sensors (all grounds connected together)

Upgrading from the previous version

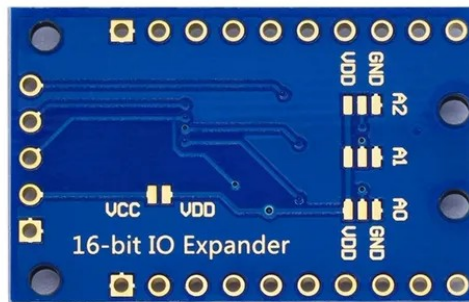
The collective sensor moved from **A0** to **A1**, and pin **A2** is new (twist throttle). Move that wire when you update an existing collective, otherwise the lever won't respond.

2.2.- 360° pogo connector

All buttons reach the base through the four pins of the 360° pogo connector: **VCC, GND, SDA and SCL**. The wire order must match exactly on the grip side and the handle side. A swapped pair is the most common cause of a grip that is not detected.

2.3.- PCF8575 button expanders (grip)

- **Expander #1 – 0x20 (buttons 1–16):** leave it as it came from the factory.
- **Expander #2 – 0x21 (buttons 17–32):** on the back of the board, bridge pad **A0** to **VDD**.



PCF8575 address pads (A0, A1, A2)

Each button is wired between its expander pin and GND. Appendix B lists the pin of every button.

3.- Hardware verification (I2C scan)

1. Open `I2C_Scanner.ino` (in the FIRMWARE folder of the official repository, section 4.5) in the Arduino IDE.
2. Select **Tools** → **Board** → **Arduino Leonardo** and the collective's port.
3. Upload the sketch with the grip **attached**, and open the Serial Monitor at **115200** baud.
4. The output must list: **0x20 and 0x21**.

Result	Likely cause
No addresses at all	The grip is not connected, the pogo connector wires are in the wrong order, or SDA/SCL are swapped.
Only 0x20 appears	The A0 bridge on expander #2 is missing, or both expanders share 0x20.
Only 0x21 appears	Expander #1 has a bridge it should not have, or a loose wire on it.

The two axis sensors are analog and do not appear in the scan; check them later on MON
 › INPUT.

4.- Firmware installation and updates

4.1.- Required libraries

Header	Library	How to install
Joystick.h	ArduinoJoystickLibrary (Matthew Heironimus)	Download the ZIP from GitHub → Sketch → Include Library → Add .ZIP Library
PCF8575.h	PCF8575 (Rob Tillaart)	Library Manager
Wire.h, EEPROM.h	Built in	—

Note

Several libraries share the name "PCF8575". Install the one by **Rob Tillaart**; the others use a different API and the sketch will not compile.

4.2.- Custom USB name and VID/PID (do this before the first upload)

By default the collective shows up in Windows and in games as "Arduino Leonardo". Giving it its own name and USB IDs makes it easy to recognize, and keeps games from mixing up its bindings with other Arduino-based devices. Set them **before** uploading, so you only flash once.

1. Find the AVR core folder that contains `boards.txt`, usually `%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\avr\1.x.x\` (the exact path may vary).
2. In that same folder, create a text file named `boards.local.txt` with the lines below. Values in this file override `boards.txt` without editing it.
3. Restart the Arduino IDE.

```
leonardo.build.vid=0x2342
leonardo.build.pid=0x803A
leonardo.build.usb_product="Black Shark Collective"
```

If you also build other Noisy Boi devices

`boards.local.txt` applies to every Leonardo / Pro Micro you upload from this PC. Each device needs its own PID (for example, 0x8038 for the joystick, 0x8039 for the throttle and 0x803A for the collective), so edit the file before flashing a different device.

4.3.- Upload the firmware

1. Open `03_Collective.ino`. Always use the latest version from the FIRMWARE folder of the official repository (section 4.5).
2. Select **Tools** → **Board** → **Arduino Leonardo** and the collective's port.
3. Click **Upload**.
4. If the upload fails because the port disappears, short **RST** to **GND** on the Pro Micro and click Upload right away (some clones need two quick taps).

After the upload, the collective shows up in Windows as a game controller (2 axes, 32 buttons) **and** as a serial (COM) port. The app uses that port. You no longer need to edit the code with calibration values, or remove the serial code after calibrating.

4.4.- Factory defaults

Setting	Default
Inverted axes	None
Alpha 1 (collective, Rx)	0.35
Alpha 2 (twist throttle, Ry)	0.35
Jitter threshold	15
Calibration	Generic values; calibrate before flying (section 5.5)

4.5.- Firmware updates

Firmware updates, fixes and new features are published in the official GitHub repository:

<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main>

1. Open the repository and download the latest collective firmware from the FIRMWARE folder (**Code** → **Download ZIP**, or open the file and use **Download raw file**).
2. Check the repository notes for changes, new required libraries or new steps before updating.

3. Close the Tactical MFD Control Hub (it uses the same COM port), then upload the new `03_Collective.ino` exactly as in section 4.3. Check that `boards.local.txt` still has the collective's PID.
4. Reconnect with the app, press **STATUS** on the LINK page to see the installed version in the log (line Firmware: ...), and check CTRL > INV / FILT and MON > INPUT.

Your settings are kept

Calibration and settings live in EEPROM, so a normal update keeps them. Recalibrate only if the update notes ask for it, or if the axes behave differently afterwards.

Watch the repository

On GitHub, click **Watch** → **Custom** → **Releases** to get an email when a new version is published.

5.- Tactical MFD Control Hub (configuration app)

The Tactical MFD Control Hub is a Windows desktop app styled as a fighter-jet multifunction display (MFD). With the collective you can use it to:

- calibrate the collective lever and the twist throttle,
- invert axes and tune the filters,
- check both axes and all 32 buttons.

All settings are stored on the collective itself. **You do not need the app running to fly.**

5.1.- Installation and launch

The app is a single Windows program (Windows 10 or 11, 64-bit). It needs no installer and no additional software.

1. Download `Tactical MFD Control Hub.exe` from the **Releases** page of the official repository (<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/releases>) and save it in any folder, for example `Documents\HOTAS`. You can create a desktop shortcut to it.
2. Double-click it to start. The first start can take a few seconds while Windows unpacks the program.
3. If Windows shows "**Windows protected your PC**", click **More info** → **Run anyway**. The app is not signed with a paid code-signing certificate, so Windows SmartScreen warns about any new program. This happens only the first time.
4. If your antivirus quarantines the file, restore it and add an exception for it. Small independent .exe tools are sometimes flagged as a false positive.

Before connecting

Only one program can use the COM port at a time. **Close the Arduino IDE Serial Monitor** (and any other serial tool) before connecting from the app.

5.2.- Navigating the MFD

Like a real MFD, the screen is surrounded by Option Select Buttons (OSBs). The label next to each OSB shows what it does on the current page. You can click the OSBs, or click directly on the rows, toggles and sliders shown on the screen.

OSB group	Function
Bottom row	Page selection: LINK , CTRL , MON . The active page is boxed.
Left column	Sub-pages (or the port list on LINK).
Right column	Page actions. On CTRL: UP / DN move the cursor, - / + change a slider, SEL toggles or runs the selected row.

The **LINK** flag in the header strip lights up while the app is connected. The GUN, MSL, TD and HFB flags belong to the joystick's haptic features and stay off with the collective.

5.3.- LINK page – connecting to the collective



LINK page: port list, connection state and serial log

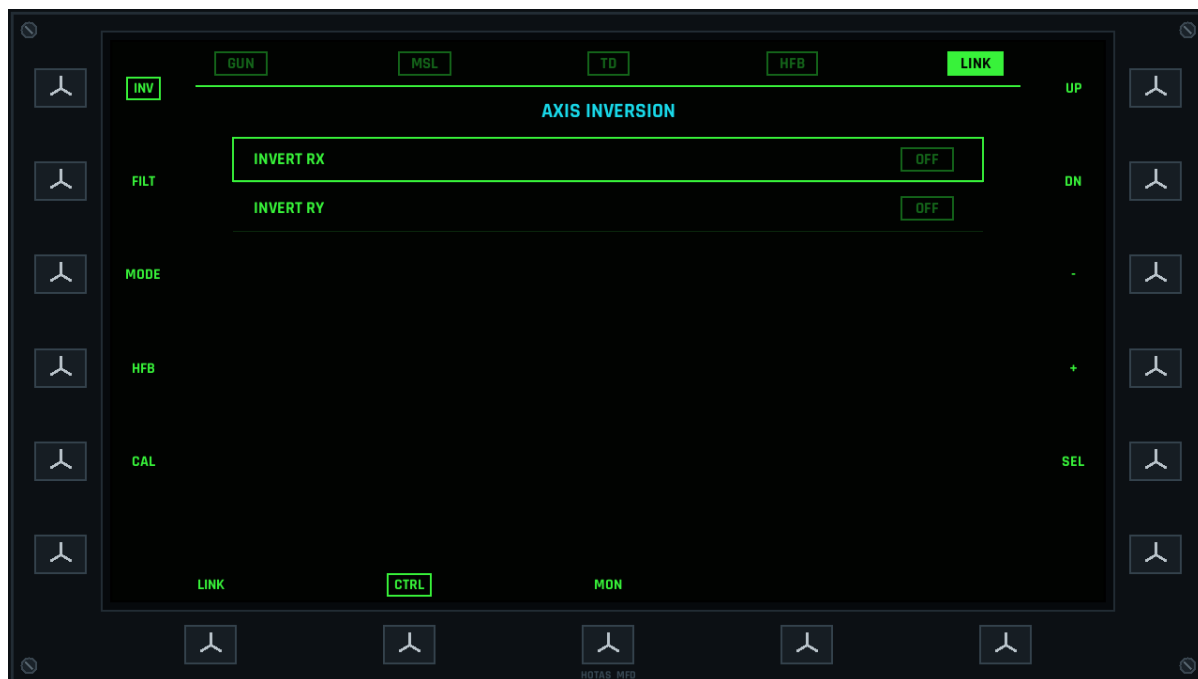
1. Plug in the collective and press **SCAN** (bottom-left OSB) to refresh the port list.
2. Select the collective's COM port with the left OSBs (or click it in the list).
3. Press **CONN**. The app requests the device status.

4. When the collective answers, **DEVICE** shows **COLLECTIVE** and the app opens the CTRL page automatically.
 - **STATUS** asks the collective for its current settings again, including the stored calibration ranges.
 - **DISC** closes the connection.
 - Not sure which port it is? Note the list, unplug the collective, press SCAN, and see which port disappeared.

5.4.- CTRL page – collective settings

Choose a sub-page with the left OSBs: **INV**, **FILT**, **CAL**. Every change is sent to the collective and saved immediately; the app then re-reads the status to confirm. The collective has no special modes (MODE shows "NO SPECIAL MODES FOR THIS DEVICE") and no haptics (HFB is for the joystick only).

INV – Axis inversion



Toggle **INVERT RX** (collective lever) or **INVERT RY** (twist throttle) until each control moves the right way in-game. For example, pulling the collective up should raise the value.

FILT – Filters



CTRL > FILT

Setting	Range	Default	Effect
ALPHA 1 (COLLECTIVE RX)	0.01 – 1.00	0.35	Collective lever filter. Lower = smoother but slower to respond. Higher = more direct, more noise.
ALPHA 2 (TWIST RY)	0.01 – 1.00	0.35	Same, for the twist throttle.
JITTER THRESHOLD	0 – 500	15	Minimum change (output range ± 16384) before an axis value updates. Raise it if an axis flickers at rest.

Fast movements are not slowed down: when a control moves quickly the filter adapts and follows it almost instantly. Drag a slider and release it, or use **- / +**; the value is sent when you release.

5.5.- CAL – Calibration

Calibration teaches the firmware the real minimum and maximum of each sensor, so both axes use their full range. Calibrate once after the build, and again if you change a sensor, a magnet or the springs. Swapping grips does not require recalibration.

1. Connect on the LINK page, then go to **CTRL > CAL**.
2. Select **START CALIBRATION WIZARD**. The collective starts with the first axis right away.

3. **Rx – collective lever:** move it slowly from fully down to fully up, several times, until the **min / max** values stop changing. Select **SAVE AXIS (NEXT)**.
4. **Ry – twist throttle:** turn it from one stop to the other, several times. Select **SAVE AXIS (NEXT)**.
5. The screen shows **CALIBRATION COMPLETE!** and the values are stored.



Calibrating the collective lever: live min/max values

Important

The app refuses to save a range that is too small (less than 20 counts): move the control through its full travel and try again.

While the wizard runs, the collective stops sending inputs to the PC. Finish it before flying.

If the app is closed in the middle of a calibration, the collective keeps waiting. Reopen the app and reconnect (it picks up the wizard where it was), or unplug and replug the USB cable to abort.

5.6.- MON page – monitoring

INPUT – Collective inputs



MON > INPUT: axes and buttons

Both axes and all 32 buttons are shown live. Use **JOY-** / **JOY+** to choose the collective and **SCAN** to refresh the list. A0 is the collective lever (Rx) and A1 the twist throttle (Ry). The app counts buttons from **0**: button 0 in the app is button 1 in the inputs layout. This is the quickest way to check a newly swapped grip.

STAT – Status summary



MON > STAT

A one-page summary of the connection and the device settings. Handy when you ask for support: send a screenshot of this page.

6.- Troubleshooting

Symptom	What to check
App: "Failed to connect"	Another program has the port open (Arduino Serial Monitor, another app instance). Close it and press CONN again.
Connected, but DEVICE stays "___"	Press STATUS. If nothing appears in the log, re-flash the firmware and check that you selected the collective's COM port.
No buttons respond	The grip is not detected: check the pogo connector wire order and run the I2C scan.
Buttons 17–32 don't respond	Expander #2 is not at 0x21: check the A0 bridge.
Collective lever doesn't move after an upgrade	The sensor is still on A0: move it to A1.
Twist throttle doesn't move	Check the A2 wire, the SS49E power and the position of the two 30×10×4 mm magnets.
An axis jitters at rest	Raise the JITTER THRESHOLD, or lower the corresponding ALPHA.

Symptom	What to check
Axis moves the wrong way	CTRL > INV.
Axis doesn't reach its ends, or has a dead zone at one end	Recalibrate using the full travel of both controls.
Inputs freeze in-game	A calibration is still running on the collective: finish it, or replug the USB cable.
Games confuse the collective with another device	Give it its own PID (section 4.2) and rebind it in the game.

Appendix A.- Serial command reference

Everything the app does can also be done from any serial terminal (for example, the Arduino Serial Monitor at **115200** baud with **Newline** line ending). Commands are not case-sensitive.

Command	Action
STATUS	Print the current settings and calibration ranges
CAL	Start the calibration: Rx (collective) then Ry (twist throttle). For each axis, move it end to end and press Enter (empty line) to save.
INV_RX, INV_RY	Toggle the inversion of that axis
FIL1 <0.01–1.00>	Set Alpha 1 (collective filter)
FIL2 <0.01–1.00>	Set Alpha 2 (twist throttle filter)
JITTER <0–500>	Set the jitter threshold

Appendix B.- Button to expander pin map

Button numbers match the inputs layout in the Assembly Instructions. Wire each button between the listed pin and GND.

Button	Expander · pin	Button	Expander · pin
1	#1 (0x20) · P00	17	#2 (0x21) · P00
2	#1 (0x20) · P01	18	#2 (0x21) · P01
3	#1 (0x20) · P02	19	#2 (0x21) · P02
4	#1 (0x20) · P03	20	#2 (0x21) · P03
5	#1 (0x20) · P04	21	#2 (0x21) · P04
6	#1 (0x20) · P05	22	#2 (0x21) · P05
7	#1 (0x20) · P06	23	#2 (0x21) · P06
8	#1 (0x20) · P07	24	#2 (0x21) · P07
9	#1 (0x20) · P10	25	#2 (0x21) · P10
10	#1 (0x20) · P11	26	#2 (0x21) · P11
11	#1 (0x20) · P12	27	#2 (0x21) · P12
12	#1 (0x20) · P13	28	#2 (0x21) · P13
13	#1 (0x20) · P14	29	#2 (0x21) · P14

Button	Expander · pin	Button	Expander · pin
14	#1 (0x20) · P15	30	#2 (0x21) · P15
15	#1 (0x20) · P16	31	#2 (0x21) · P16
16	#1 (0x20) · P17	32	#2 (0x21) · P17

For questions contact me at: the.noisy.boi@gmail.com

ALL DONE – HAPPY FLYING!